POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor and Computer Architecture

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Difference between Von-Newmann and Harvard architecture with necessary diagram. 7

   b) What are the different types of FLAGS used in 8085 microprocessor? Show the status for the Flags after performing the following operation 5F H+ 02 H. 8

2. a) Explain the timing diagram for ORI 07H. 8

   b) Write an ALP for the following addition. 7

   $1^2+2^2+3^2+4^2+5^2+6^2+7^2+8^2+9^2$

3. a) What are the addressing modes of 8086 microprocessor? Explain in detail. 7

OR

Explain ALP development tools in 8086 with examples.

   b) Design a very simple CPU with given instruction set: 8

Instruction	Instruction Code	Micro-operations
ADD	00 AAAAAA	AC← AC+M[AAAAAA]
JMP	01 AAAAAA	PC← AAAAAA
OR	10 AAAAAA	AC← AC^{V}M[AAAAAA]
SKIP	11 XXXXXX	PC← PC+1

Write down the RTL codes for the CPU, draw state diagram, and register section and ALU.

4. a) Design a hardwired control unit for the CPU given in question number 3b. 7

   b) Design a micro-sequencer control unit with horizontal microcode for the given CPU in question number 3. b. 8

5. a) Draw the flowchart for the Booth's algorithm and perform -9×-9. 7

OR

Show the division of the number 14/2 using Restoring Division algorithm.

b) What is memory hierarchy? Differentiate between direct, associative and set associative mapping. 8

6. a) Draw and explain the flow of Interrupt I/O during input/output transfer. How is it more efficient than programmed I/O? 7

b) What are the major conflicts occurring due to instruction pipelining in RISC? Explain. Describe the solutions to correct data conflicts. 8

7. Write short notes on: **(Any two)** 2×5

a) Flynn's taxonomy

b) HDMI

c) Instruction set